

NEURAL SPACETIME SEARCH WITH GRTECLYN: GEOMETRY-FIRST AND MATTER-FIRST BASELINES

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We present a compact baseline for automated searches over dynamical spacetime initial data. A Python optimizer proposes low-dimensional parameters, **GRTEclyn** evolves each candidate on GPU, and diagnostics score constraint health, operational faster-than-light (FTL) benefit, stability, and energy-condition cost. The paper separates two regimes. In the *geometry-first* regime the optimizer deforms the metric directly and uses the constraints to infer the required matter. This is cheap and has already found a standing directional shortcut, but it does not supply the momentum source needed for transport. In the *matter-first* regime the optimizer proposes scalar-matter configurations and **GRtresna** solves the elliptic constraints before **GRTEclyn** evolves the result. This is the right path toward physical searches, but current mixed exotic-lump MAP-Elites results require a stricter post-load constraint gate because the matter stress tensor solved by **GRtresna** need not equal the stress tensor evolved by **GRTEclyn**. We therefore treat the matter-first campaign as a promising engineering milestone, not yet as a validated FTL metric discovery.

I. GOAL AND SEARCH LOOP

Most warp and wormhole metrics are hand-designed static ansätze. Here the metric is instead treated as the output of a black-box search over dynamical 3D initial data. Throughout this paper “FTL” means an effective shortcut through curved spacetime relative to a flat-spacetime control; it does not imply local superluminal motion outside the local light cone.

The search problem is deliberately operational. A useful candidate must be non-trivial, approximately constraint-satisfying, dynamically stable for more than a short smoke window, and faster than a flat control according to a probe that is not merely reading off a gauge artifact. The loop is

$$\theta \rightarrow \text{initial data} \rightarrow \text{GRTEclyn} \rightarrow \text{diagnostics} \rightarrow S(\theta),$$

where S is a bounded scalar score used by CMA-ES or MAP-Elites. Health terms are multiplied by a non-triviality gate so flat space cannot win simply because it is stable.

The score is not a single visual heuristic. It combines four classes of diagnostics. First are shortcut indicators: axial null travel time F_{null} , proper-distance compression, throat pinching, and Natário–Alcubierre expansion asymmetry. Second is the operational probe F_{op} , an arrival-time comparison against a flat control on the same grid. Third are health terms: survival, Hamiltonian/momentum constraints, lapse collapse, trapped-surface flags, and growth of χ , K , and constraint norms. Fourth are risk terms: required negative energy, matter/effective energy conditions, and tidal/curvature proxies. The exact weights are not essential for this paper; the key point is that a candidate must be both a shortcut and a healthy evolved spacetime.

For an axial null probe on a conformally flat slice,

$$T_{\text{curved}} = \int_{-L}^L \frac{dx}{\alpha(x)\sqrt{\chi(x)} - \beta^x(x)}, \quad (1)$$

$$F_{\text{null}} = \max\left(0, \frac{2L - T_{\text{curved}}}{2L}\right).$$

The final operational test compares arrival time to a flat control,

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \quad (2)$$

and is accepted only with bounded constraints and no trapped-surface crossing. The risk side tracks Hamiltonian/momentum norms, collapse indicators, geometry growth, energy-condition margins, and required exotic energy

$$\rho_{\text{req}} = \frac{R + K^2 - K_{ij}K^{ij}}{16\pi}. \quad (3)$$

A non-triviality gate suppresses flat Minkowski space: health rewards are counted only when the geometry has a real shortcut, curvature signal, asymmetry, or exotic content. This prevents the optimizer from selecting “nothing happens” as the most stable solution.

II. GEOMETRY-FIRST SEARCH

The first baseline asks the optimizer to propose geometric and gauge fields directly. For the conformally flat recipe, the Hamiltonian and momentum constraints are read backwards:

$$\rho = \frac{R + K^2 - K_{ij}K^{ij}}{16\pi}, \quad j_i = \frac{1}{8\pi} D_j (K_i^j - \delta_i^j K). \quad (4)$$

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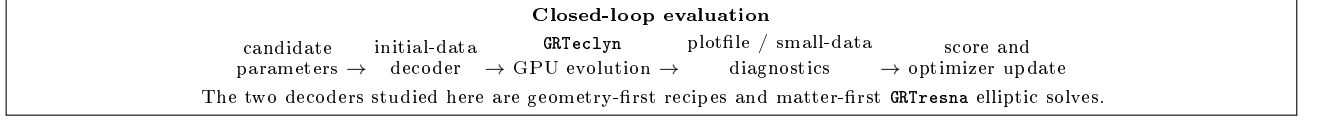


FIG. 1. **Search loop.** The optimizer sees only the scalar objective; physics enters through the initial-data map and through post-evolution diagnostics.

Regions with $\rho < 0$ become an exotic-matter cost. In the simplest radial recipe,

$$q(r) = q_\infty + \sum_n a_n B_n(r), \quad (5)$$

where q denotes χ , lapse, shift, or K . Angular modes are then added in the gauge sector. This is computationally cheap because it avoids a 3D elliptic solve, but it has a hard limitation: with $K = \text{const}$ and $\Pi = 0$ the momentum density vanishes. The recipe can bend space into a lens, but it cannot provide a co-moving matter current.

A. Validated Baseline Result

The best geometry-first outcome is a standing Alcubierre-like dipole. The search found non-spherical gauge/geometric structure with a localized exotic core rather than a translating bubble. A representative promoted candidate, `eval_000188`, was replayed to $t = 16$ at $N_{\text{full}} = 192$. It retained a weak evolved operational shortcut,

$$F_{\text{op}}^{\text{ev}} = 5.6 \times 10^{-3},$$

with decreasing constraint norms $\|H\|_{L^2} : 1.1 \times 10^{-3} \rightarrow 4.2 \times 10^{-4}$ and final $\|M\|_{L^2} = 3.6 \times 10^{-5}$. The required negative energy stayed localized,

$$\int \rho_- dV \simeq 0.13, \quad \min \rho_{\text{req}} \simeq -10^{-3}.$$

This is a useful automated-discovery baseline: the search can identify a structured, low-exotic shortcut geometry that survives longer than the smoke window. It is not a transport metric. The structure stands and relaxes rather than carrying a passenger region across the grid.

The lesson is clear. Geometry-first search is a fast way to discover useful geometric motifs and to test scoring diagnostics, but the optimizer naturally prefers a cheap stationary lens over a harder translating bubble. A transport metric needs a sustained momentum density j_i , because $G_{0i} = 8\pi j_i$ ties matter current to the frame-dragging shift.

III. MATTER-FIRST SEARCH

The matter-first route inverts the map. The optimizer proposes matter; an elliptic solver returns the compatible

`eval_000188` validated ($N = 192$, $S = 16.75$): the warp is supported by a localized exotic core; ordinary matter is negligible ($\sim 20\text{--}150\times$ weaker, outgoing-wave level)

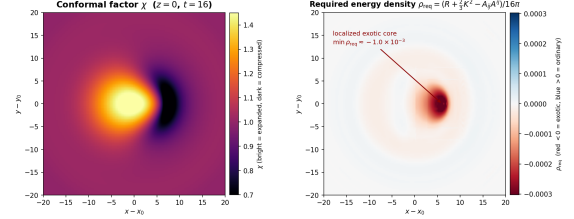


FIG. 2. **Geometry-first dipole.** The validated candidate is a static exotic lens: the compression lobe in χ is supported by a localized negative required energy density.

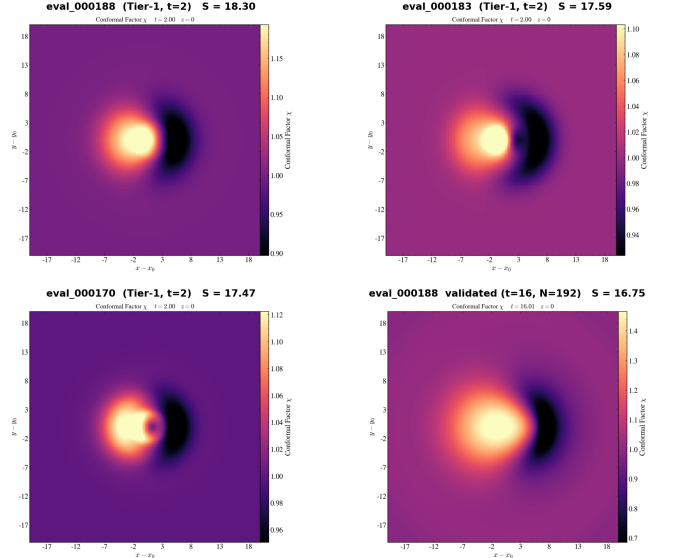


FIG. 3. **Directional geometry-first candidates.** The top candidates share the same standing expand-behind/compress-ahead polarity. This is the geometry a warp channel needs, but not the matter current needed to move it.

geometry. For scalar matter,

$$\rho = \frac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \frac{1}{2}\Pi^2 + V(\phi), \quad S_i = -\Pi \partial_i \phi. \quad (6)$$

GRTresna solves the Hamiltonian and momentum constraints for the conformal factor and longitudinal extrinsic curvature, writes a `.gridinit` file, and GRTeclyn evolves it. This adds the missing ingredient from Sec. II: searchable momentum-carrying matter.

Matter-first discovery stages

16D shell vector $\rightarrow$ MAP-Elites archive $\rightarrow$ optional CMA-ES refinement $\rightarrow$ high-resolution promotion / validation
 The shell vector expands to five Fibonacci-distributed scalar clouds. MAP-Elites keeps diverse families by descriptor cell instead of collapsing to one scalar optimum.

FIG. 4. **Shell discovery workflow.** This is the compact form of the README pipeline: broad quality-diversity search, optional local refinement, and fresh high-resolution replay.

Per-candidate matter-first loop

sample shell params $\rightarrow$ GR_{Tresna} solve $\rightarrow$ Ham/Mom gate $\rightarrow$.gridinit $\rightarrow$ GR_{Teclyn} evolve $\rightarrow$ score/archive
 The missing production gate is a post-load GR_{Teclyn} constraint check using the matter model that is actually evolved.

FIG. 5. **Matter-first evaluation.** GR_{Tresna} convergence is necessary but not sufficient: the loaded state must also satisfy the constraints under the GR_{Teclyn} evolution stress tensor.

A. Shell Parameterization and MAP-Elites

The current discovery ansatz uses a 16-dimensional shell vector. It places a small number of scalar clouds on a Fibonacci lattice over the sphere, rotates the lattice by a searched orientation axis, and assigns toroidal, poloidal, and radial velocity components. Low-order dipole/quadrupole modulations control matter asymmetry, while an exotic wedge selects where negative-energy support is attempted. MAP-Elites is used because preserving diverse mechanism families is more valuable here than only optimizing a single scalar score.

The recent shell campaign produced strong operational signals in the search logs, including promoted candidates with large $F_{\text{op}}^{\text{ev}}$ after longer replay. These are important engineering signals: the optimizer, elliptic solver, GPU evolution, plotfile consumption, and archive machinery can all run at scale. They are not yet a validated physical result for the mixed exotic-lump search space.

B. Current Trust Issue

The reason is a matter-model mismatch. GR_{Tresna} may solve a source that is not the same stress-energy GR_{Teclyn} evolves after loading .gridinit. Three effects matter.

First, exotic lumps in GR_{Tresna} enter the source as independent phantom components, with their kinetic and momentum terms multiplied by a negative sign. The written ϕ and Π fields do not carry that sign. If GR_{Teclyn} evolves the combined field as a canonical scalar, the local Hamiltonian source has the wrong sign in exotic regions.

Second, the solved source treats lumps as independent fields and drops inter-lump cross terms. The evolved single scalar contains the combined field, so $|\nabla \sum_k \phi_k|^2$ includes cross terms wherever clouds overlap.

Third, the solver source can include a mass potential while the current GR_{Teclyn} default potential is zero. Thus a small GR_{Tresna} Hamiltonian residual certifies the elliptic solve, not necessarily the state that is actually

evolved.

The practical criterion is therefore stricter:

A matter-first candidate is trusted only if the loaded GR_{Teclyn} state passes the Hamiltonian and momentum constraints at $t = 0$ using the same matter model that will be evolved, and then remains bounded under replay.

Until that gate is added and applied to the shell archive, the MAP-Elites numbers should be reported as promising search signals rather than as evidence for a validated FTL geometry.

IV. PATH TO A TRUSTWORTHY MATTER-FIRST CAMPAIGN

The next campaign should be organized around falsification rather than score maximization. The required steps are short and concrete.

a. Post-load constraint gate. After GR_{Tresna} writes .gridinit, load the file into GR_{Teclyn}, compute H and M_i with the actual evolution matter model, and reject candidates before GPU evolution if the $t = 0$ residual is large. This is the missing gate in Fig. 5.

b. Representable matter. Either restrict the search to matter that a single evolved scalar can represent (for example all-canonical or all-phantom, matched potential, limited overlap), or promote the evolution model to the same independent-field stress tensor that GR_{Tresna} solved. A frozen prescribed source should not be used for physics claims, because it repeats the known failure mode $G_{ab} \neq 8\pi T_{ab}$ as soon as the geometry evolves.

c. Transport objective. The current objective rewards the existence of a shortcut on a slice. A warp metric needs a co-moving worldtube: a localized passenger region moving a finite distance while the surrounding geometry is approximately stationary in its own frame. The score should require this condition, otherwise the optimizer will continue to prefer static lenses.

d. Resolution and observer checks. Any surviving candidate must be replayed on a resolution ladder and evaluated with observer-robust energy conditions, not only matter-sector diagnostics. The target validation ladder is: constructed, non-trivial, post-load constraint-clean, operational, persistent, observer-robust, resolution-converged, and analytically extractable.

e. Analytical extraction. A numerical elite is only a coefficient vector. The final step is fitting a closed-form metric or matter profile, re-solving the constraints from that form, and replaying it. This is what turns the search result into a checkable spacetime rather than an isolated grid artifact.

V. CONCLUSION

The geometry-first search is a useful baseline: it is fast, debugs the scoring pipeline, and has found a clean stand-

ing shortcut supported by a localized exotic core. Its limitation is also decisive: it does not supply the sustained momentum source needed for transport. Matter-first search with **GRTresna** is the better route because it exposes momentum-carrying matter and solves the constraints before evolution. However, the current mixed exotic-lump **MAP-Elites** results should be treated conservatively until the post-load **GRTeclyn** constraint gate and matched matter model are in place. The near-term goal is therefore not to claim a new FTL metric, but to turn the matter-first pipeline into a falsifiable and constraint-clean discovery engine.

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